Lecture 6

Homology and the Euler characteristic

Objectives. *Homology distinguishes closed chains from chains that bound higher-dimensional pieces. A decomposition into triangles makes this distinction a calculation with boundary matrices. We compute these matrices for the torus and the projective plane. Their alternating dimensions give the Euler characteristic. On a periodic lattice, the same quotient describes classes of closed strings in the toric code.*

By the end of this lecture you will be able to:

- Construct boundary matrices from ordered simplices.
- Compute cycles, boundaries and homology with specified coefficients.
- Prove the Euler characteristic formula for a finite complex.
- Identify the two independent binary string classes on a torus.

6.1 Simplices and their identifications

A *k-simplex* is the convex hull of $k+1$ affinely independent points, with a chosen ordering of these vertices. The standard simplex is

$$\Delta^k = \{(t_0, \dots, t_k) \in \mathbb{R}^{k+1} : t_i \geq 0,\ t_0 + \cdots + t_k = 1\}.$$

Deleting one vertex gives a *face*, with its inherited order. A point, an interval and a triangle are simplices of dimensions zero, one and two. The order of the vertices orients an edge from the first vertex to the second. It also specifies the signs of higher-dimensional faces.

A Δ-*complex structure* on X is a collection of continuous maps $\sigma\colon \Delta^k \to X$, called its simplices, with the following properties. Each map is injective on the interior of its domain. These interiors have disjoint images and together cover X. The restriction to each face is a designated lower-dimensional simplex, using the affine identification preserving the vertex order. The topology is specified by these maps: a subset $A \subseteq X$ is closed exactly when $\sigma^{-1}(A)$ is closed for every simplex σ. Thus the space is obtained by attaching the faces in the specified way.

Different vertices of one domain simplex may have the same image in X. An edge can therefore become a loop. By contrast, in a *simplicial complex* each simplex has distinct vertices in the space and any two simplices intersect in a common face or not at all. The torus model below has one vertex and two triangles. It is a Δ-complex, not a simplicial complex. Vertex names such as $[v_0, v_1, v_2]$ refer to ordered domain vertices even when some images coincide.

6.2 Chains and the boundary map

Fix a *coefficient ring* R. Our default is $R = \mathbb{Z}$. The alternatives used here are $\mathbb{Z}_2$, the field with arithmetic modulo two, and the real field $\mathbb{R}$. Integers form a ring, not a field. A *k-chain* is a finite formal sum of the k-simplices with coefficients in R. These sums form the *chain group* $C_k(X; R)$. Chains are added and multiplied by coefficients componentwise. Every chain has a unique expression in the simplex generators. When R is a field, they form a vector-space basis. Write $C_{-1} = 0$.

The *boundary map* $\partial_k \colon C_k(X; R) \to C_{k-1}(X; R)$ is the linear map defined on a simplex by alternating deletion:

$$\partial_k[v_0, \ldots, v_k] = \sum_{i=0}^{k} (-1)^i [v_0, \ldots, \widehat{v_i}, \ldots, v_k] \quad (k \geq 1), \qquad \partial_0 = 0.$$

The hat means that the vertex is omitted. Each face is read as the designated simplex of the complex. This ∂ acts on chains; it is not a coordinate derivative.

For an ordered triangle, the formula is

$$\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1].$$

The negative sign reverses the contribution of the middle edge. Over $\mathbb{Z}_2$ the negative sign equals a positive sign. Repeated occurrences of an edge still have to be added in the chosen ring.

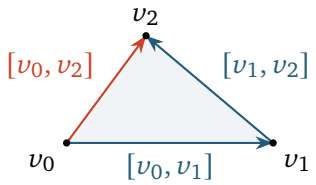


Figure 6.1: The ordered edges point from lower to higher vertex index. The boundary traverses the middle edge with a negative coefficient.

Theorem 6.1 (The boundary of a boundary)**.** *For every $k \geq 1$, the composition $\partial_{k-1}\partial_k$ is zero.*

Proof. For $k = 1$ the target boundary is $\partial_0 = 0$. For $k \geq 2$, fix indices $i < j$ in an ordered k-simplex. Deleting v_i first moves v_j to position $j - 1$. The resulting face has coefficient $(-1)^i(-1)^{j-1}$. Deleting v_j first leaves v_i in position i and gives coefficient $(-1)^j(-1)^i$. These are opposite signs. Every face with two deleted vertices occurs in exactly this pair. The pairs cancel before the face identifications, so they also cancel after those identifications. Linearity proves the assertion for every chain. ∎

For the triangle, applying the boundary once more gives

$$(v_2 - v_1) - (v_2 - v_0) + (v_1 - v_0) = 0.$$

Here a vertex name denotes its generator in C_0. A sequence of groups linked by homomorphisms whose consecutive compositions vanish is called a *chain complex*.

6.3 Cycles, boundaries and coefficients

A *cycle* is a chain with zero boundary. A *boundary* is a chain that is the boundary of a chain one dimension higher. Their groups and the resulting *homology group* are

$$Z_k(X;R) = \ker \partial_k, \qquad B_k(X;R) = \operatorname{im} \partial_{k+1}, \qquad H_k(X;R) = Z_k(X;R)/B_k(X;R).$$

The boundary-of-boundary theorem gives $B_k \subseteq Z_k$, so the quotient is defined. Two cycles are *homologous* when their difference is a boundary. Homology records which cycles remain after this identification. The symbols Z_k and B_k denote groups, distinct from the coefficient ring $\mathbb{Z}$ and the braid group B_n of Lecture 5.

For example, take the boundary of a triangle, with no filled face. The sum of its cyclically directed edges is a cycle. A general cycle has equal coefficients on those three directed edges, since its boundary vanishes at each vertex. There are no two-dimensional chains, so this cycle generates $H_1 \cong R$. Filling the triangle makes that same sum a boundary and gives $H_1 = 0$.

For a finite complex, order the simplices of each dimension. The boundary matrix D_k has a column for each source k-simplex and a row for each target $(k-1)$-simplex. Its entries are the signed face coefficients. Write c_k for the number of k-simplices and r_k for the rank of D_k over a specified field. Rank–nullity and the dimension formula for a quotient, recalled from *Linear Algebra for Physics*, give

$$\dim H_k(X;R) = (c_k - r_k) - r_{k+1} \qquad \text{when } R \text{ is a field.}$$

Over $\mathbb{Z}$, integer relations determine an abelian-group quotient instead. Real row reduction alone does not determine that group.

An element is a *torsion element* if it is nonzero and some positive integer multiple is zero. For an algebraic example, take the chain complex with $C_2 = C_1 = \mathbb{Z}$, boundary ∂_2 given by multiplication by two, and $\partial_1 = 0$. Its first homology is $\mathbb{Z}/2\mathbb{Z}$. With real coefficients, multiplication by two is onto, so the first homology is zero. With $\mathbb{Z}_2$ coefficients it is the zero map, so the first homology is $\mathbb{Z}_2$. This illustrates why coefficients are part of the answer. For a finite complex, the *kth Betti number* means $b_k = \dim_{\mathbb{R}} H_k(X;\mathbb{R})$. Real coefficients discard torsion information.

6.4 The Euler characteristic

For a finite Δ-complex of dimension d, its *Euler characteristic* is the alternating simplex count

$$\chi(X) = \sum_{k=0}^{d} (-1)^k c_k.$$

For a triangulated surface this is vertices minus edges plus triangles. The symbol χ denotes a topological count here, not the character of Lecture 5.

Theorem 6.2 (Euler characteristic and homology). *For a finite Δ-complex and any field R, $\chi(X) = \sum_{k=0}^{d} (-1)^k \dim_R H_k(X;R)$. In particular, $\chi(X) = \sum_k (-1)^k b_k$.*

Proof. Set $r_0 = r_{d+1} = 0$. The dimension formula for homology gives $c_k = \dim H_k + r_k + r_{k+1}$. On summing with alternating signs, the rank terms cancel:

$$\sum_{k=0}^{d} (-1)^k c_k = \sum_{k=0}^{d} (-1)^k \dim H_k + \sum_{k=0}^{d} (-1)^k r_k + \sum_{k=0}^{d} (-1)^k r_{k+1} = \sum_{k=0}^{d} (-1)^k \dim H_k.$$

Indeed, the coefficient of each r_j with $1 \le j \le d$ is $(-1)^j + (-1)^{j-1} = 0$. The endpoint ranks vanish. ∎

For a single filled triangle, the counts are 3, 3 and 1, giving $\chi = 1$. Its homology has one degree-zero generator and no higher generators, which gives the same alternating sum.

6.5 Invariance and closed surfaces

The calculation uses a decomposition. Its independence of that choice requires a theorem, rather than a comparison of a few examples.

Theorem 6.3 (Homology invariance · *cited*). *The homology of a Δ-complex agrees with its singular homology. Homotopy equivalences induce isomorphisms of these groups, with the coefficient ring fixed. Consequently the Euler characteristic of a finite Δ-complex is invariant under homotopy equivalence.*

The comparison is Hatcher, Theorem 2.27. Homotopy invariance is Corollary 2.11; coefficients are treated in §2.2, pp. 153–155. Singular homology uses all continuous maps $\Delta^k \to X$ as generators, without the interior-injectivity requirement. That construction applies without choosing a decomposition. The Euler assertion follows from the theorem above, for example with real coefficients.

A *triangulation* is a presentation of a surface by triangles glued along edges. A triangulated surface is *orientable* if its triangles can be oriented so that their induced directions on every shared edge are opposite. We use as part of the cited surface classification below that every closed surface admits such a presentation and that this condition is independent of the choice. The torus has this property; the projective plane does not.

Theorem 6.4 (Closed surfaces · *cited*). *Closed connected surfaces are classified, up to homeomorphism, by their orientability and Euler characteristic. For the orientable surface Σ_g of genus $g \ge 0$, the Euler characteristic is $2 - 2g$, where $\Sigma_0 = S^2$.*

This is the classification theorem of Kosniowski, Theorem 11.3, together with the Euler counts in Hatcher, §2.2, pp. 146–147. For $g \ge 1$, the polygon model of Lecture 5 has one vertex, $2g$ edges and one face. Using such polygonal cells instead of triangles gives the same Euler characteristic by Hatcher, Theorem 2.44. Thus its count is $1 - 2g + 1$.

There is also a direct connection to Lecture 5. For any path-connected space, $H_1(X;\mathbb{Z})$ is the *abelianization* of $\pi_1(X)$: the quotient by the normal subgroup generated by all commutators $aba^{-1}b^{-1}$. This result is cited from Hatcher, Theorem 2A.1. Homology therefore forgets the order information in noncommuting loop classes. It need not recover the fundamental group.

6.6 Boundary matrices for two surfaces

We compute the torus over $\mathbb{Z}$ and then $\mathbb{Z}_2$. Take a square with corners P, Q, R', S counterclockwise from the bottom left. Pair the horizontal edges to the right and the vertical edges upward. All corners become one vertex v. Call the horizontal edge a, the vertical edge b and the diagonal $P \to R'$ edge c. The ordered triangles are $T_0 = [P, Q, R']$ and $T_1 = [P, S, R']$.

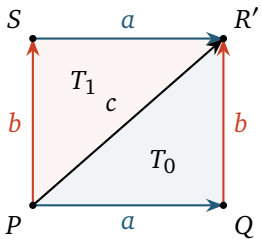


Figure 6.2: The torus has one vertex, three edges and two triangles. The upper triangle's vertex order has the opposite planar orientation to the lower one's.

Example 6.5 (The torus)**.** Both triangle boundaries are $a+b-c$. Each edge has zero boundary, since both endpoints become v. In the bases (a, b, c) and (T_0, T_1), the boundary matrices over $\mathbb{Z}$ are

$$D_1 = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}.$$

$$D_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{pmatrix}.$$

Every one-chain is a cycle. The boundaries are integer multiples of $a + b - c$. The list $(a, b, a + b - c)$ is an integer basis of C_1: its change-of-basis determinant is -1, so its inverse has integer entries. Quotienting by its last generator leaves a and b free. The kernel of D_2 consists of pairs $(m, -m)$, and there are no three-chains. The zero-dimensional quotient retains the single vertex. We have therefore obtained

$$H_0(T^2; \mathbb{Z}) = \mathbb{Z}, \qquad H_1(T^2; \mathbb{Z}) = \mathbb{Z}^2, \qquad H_2(T^2; \mathbb{Z}) = \mathbb{Z}.$$

The top generator is $T_0 - T_1$. Its signs give both triangles the same planar orientation, and their edge contributions cancel.

With $\mathbb{Z}_2$ coefficients, every entry -1 becomes 1. The ranks of D_1 and D_2 are zero and one. The homology dimensions are therefore 1, $3 - 0 - 1 = 2$ and $2 - 1 = 1$. The classes of a and b still generate H_1, and the top cycle is $T_0 + T_1$. Both coefficient choices give $\chi(T^2) = 1 - 3 + 2 = 0$.

For the projective plane, pair opposite square edges with opposite arrows. The corners P and R' become v, while Q and S become w. The edge a runs from P to S and from R' to Q. The edge b runs from P to Q and from R' to S. The diagonal c runs from P to R'.

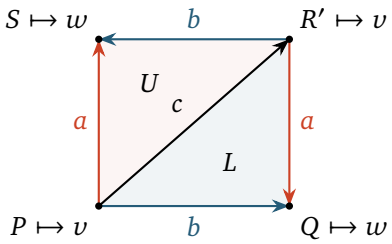


Figure 6.3: The projective plane has two vertices after identification. The diagonal c is a loop at v; both boundary-edge generators join v to w.

Example 6.6 (The projective plane over $\mathbb{Z}_2$)**.** Use the vertex basis (v, w), edge basis (a, b, c) and ordered triangles $U = [P, R', S]$ and $L = [P, R', Q]$. Work over $\mathbb{Z}_2$ throughout. Both edges a and b have boundary $v+w$, and c has zero boundary. Both triangles have boundary $a+b+c$. Thus

$$D_1 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

$$D_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Both matrices have rank one. The kernel of D_1 is generated by $a + b$ and c. Its boundary subspace is generated by $a + b + c$, so the quotient is one-dimensional, represented by c. The kernel of D_2 is generated by $U + L$. The image of D_1 identifies the two vertex classes in H_0. Consequently $H_k(\mathbb{RP}^2; \mathbb{Z}_2) \cong \mathbb{Z}_2$ for $k = 0, 1, 2$, and $\chi(\mathbb{RP}^2) = 2 - 3 + 2 = 1 = 1 - 1 + 1$. These are binary homology groups; they do not assert the same answer with integer or real coefficients.

6.7 Binary cycles and the toric code

On a square lattice with opposite sides identified, put one qubit on each edge. A qubit has basis states $|0\rangle$ and $|1\rangle$. Its phase-flip operator Z fixes $|0\rangle$ and multiplies $|1\rangle$ by -1. This operator symbol is distinct from the cycle group Z_k. A *Z-string* applies this operator on a chosen set of edges. Represent that set by a one-chain with $\mathbb{Z}_2$ coefficients: an edge has coefficient one exactly when it is selected. Adding chains means toggling the edges, since applying Z twice is the identity.

The relevant local operations are read from vertices and square faces, called *plaquettes*. A vertex condition on a string requires an even number of selected edges at that vertex. Algebraically this is $\partial c = 0$. A plaquette boundary is also a cycle. Adding it replaces a part of the string by the other edges of that face. The drawing shows this change beside two independent closed strings.

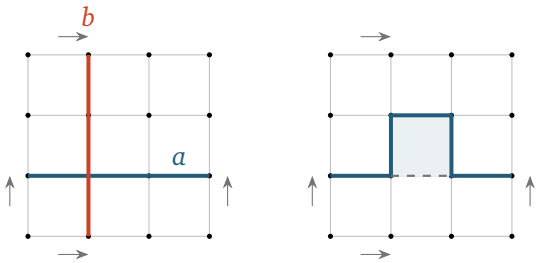


Figure 6.4: Opposite sides of each square are identified. The first panel shows the two periodic cycles. The second adds one plaquette boundary; the dashed edge is removed.

Draw every square's diagonal from lower left to upper right and order its triangles as in the torus example. A triangle boundary replaces each added diagonal in a cycle by its other two edges. Conversely, each diagonal occurs in exactly the two triangles of its square. A binary sum of triangle boundaries with no diagonal term therefore uses the two triangles of each square with equal coefficients. The two triangle boundaries add to the square boundary, so the sum is a sum of square boundaries. This proves that square-lattice

cycles modulo square boundaries agree with the homology of this triangulation. By the cited Theorem 6.3, its homology agrees with that of the two-triangle torus in Example 6.5.

Example 6.7 (Toric-code string classes · *cited; two-triangle homology proved*)**.** The binary string classes form $H_1(T^2;\mathbb{Z}_2) \cong \mathbb{Z}_2^2$. Representatives wind once in the horizontal and vertical directions. Every class is a sum of these two, so there are four classes. The identification of these lattice representatives is part of Kitaev's cited construction.

The *code subspace* is a chosen subspace of the joint qubit state space used to store quantum information. A *logical operator* preserves that subspace and acts on the states stored there. In the toric code, closed Z-strings preserve the code subspace. Plaquette boundaries act as the identity on that subspace, so homologous strings act in the same way. The two independent classes give the two logical Z operators. The full code construction supplies corresponding complementary operators and a four-dimensional code subspace: two logical qubits. This physical interpretation is cited from Kitaev (2003), §1, equations (1)–(2) and the following logical-operator argument. The homology calculation identifies the string classes; the Hilbert-space and operator assertions use that cited construction.

Self-check. Compute the Euler characteristic of the sphere from the boundary of a tetrahedron and from the boundary of an octahedron.

Answer. The tetrahedron has four vertices. Every pair is an edge, giving $\binom{4}{2} = 6$ edges, and every triple is a face, giving four triangles. The count is $4 - 6 + 4 = 2$. The octahedron has vertices at the positive and negative coordinate-axis unit vectors. Every vertex joins the four vertices other than itself and its opposite, giving $6 \cdot 4/2 = 12$ edges. A face chooses one endpoint from each axis, giving $2^3 = 8$ triangles. Its count is $6 - 12 + 8 = 2$, in agreement with the tetrahedron.

Summary. The boundary map has square zero. Homology is the quotient of cycles by boundaries, with the coefficient ring specified. Over a field, boundary-matrix ranks give its dimensions and their alternating sum is the Euler characteristic. Invariance and surface classification are cited. The torus has two independent binary one-cycle classes, which describe the two logical Z-string classes in the toric code. Lecture 7 adds smooth coordinates to spaces so that calculus can be defined on them.

Reading. Hatcher, *Algebraic Topology*, §2.1, pp. 102–107, Theorems 2.27 and 2A.1, and §2.2, pp. 146–147; Nakahara, *Geometry, Topology and Physics*, §3.3; Kosniowski, *A First Course in Algebraic Topology*, Theorem 11.3 and Chapter 29; Kitaev, *Fault-tolerant quantum computation by anyons* (2003), §1.